
Summary

AI Engineer building **production, user-facing AI systems** with a focus on turning **ambiguous business problems into scalable, decision-driven products**. Experienced in **LLM-based systems (RAG, multi-agent workflows)** and backend infrastructure for low-latency, real-world applications.

Skills

LLM Systems & Agents: Multi-Agent Workflows (LangGraph), RAG, Prompt Orchestration, Tool Use / Function Calling, Agent Memory (MCP), Guardrails, Evaluation, Observability

Machine Learning: PyTorch, TensorFlow, XGBoost, YOLO, Model Optimization, Computer Vision

Languages: Python, TypeScript, SQL, Go, C++

Backend & Infra: FastAPI, Microservices, AsyncIO, REST APIs, Docker, CI/CD (GitHub Actions), AWS, GCP, CUDA

Data & Search: Vector Databases (Pinecone, FAISS), PostgreSQL, MongoDB, ETL Pipelines

Experience

AI Software Engineer (Forward Deployed) — ProspectIntel

Jan 2026 – Present

- Building a **production AI sales intelligence platform** using agentic workflows + RAG, enabling real-time prospect research across **multiple users and concurrent queries**, significantly reducing manual analysis time.
- Owned translation of ambiguous sales workflows** into structured AI systems, driving rapid iteration and improving usability of generated insights.
- Designed **end-to-end system architecture** for agent orchestration, retrieval, and evaluation pipelines across multiple services.
- Engineered a **RAG-based intelligence pipeline** using hybrid retrieval over PostgreSQL, improving reliability from **70% to 95%** and reducing latency by **2–3×**.
- Implemented a **lazy-loading inference architecture with streaming LLM responses**, reducing perceived latency by **40%** and optimizing cost through gated model execution.
- Built **evaluation and observability pipelines (Langfuse)** to track prompt performance, latency, and token usage, reducing token consumption by **35%**.
- Optimized **retrieval and inference tradeoffs** across latency, cost, and accuracy, improving system efficiency under production constraints.

AI Software Developer — Bruise @ GMU

Jun 2025 – Present

- Engineered a **HIPAA-compliant full-stack AI platform** (FastAPI, React/Next.js) supporting real-time clinical workflows, deployed across multiple universities and used by 50+ clinicians, enabling reliable, low-latency processing of medical imaging data.
- Built a **dedicated AI inference microservice (Flask)** serving YOLO-based models with **dynamic model routing**, offloading heavy computation and improving system latency and flexibility.
- Developed a **Python-based HTTP gateway** bridging a Go backend with AI services, enabling cross-language communication and **asynchronous processing workflows**.
- Designed **data processing pipelines** to extract, clean, and normalize Base64-encoded medical images and metadata from PostgreSQL for downstream AI inference.
- Built **LLM summarization pipelines** and integrated **3D spatial mapping (Unity)**, improving clinical usability and visualization.

Software Engineer (ML) — Geoweaver (NASA / ESIP)

Feb 2024 – Jan 2025

- Built and productionized **scalable ML pipelines** for TB-scale geospatial datasets (snow water estimation, wildfire detection, ozone analysis) using Python and PostgreSQL.
- Executed and optimized workflows on **HPC systems (Hopper)**, leveraging distributed processing to improve compute efficiency by 20% and accelerate large-scale data analysis.
- Designed and integrated **multi-database support** and contributed to an in-app **code editor**, enabling flexible data access and improving developer workflows within the platform.
- Improved **UI responsiveness and feature experience**, reducing latency and contributing to a **25% increase in monthly active users**.
- Mentored a team of 4 interns in ML experimentation and system design, **co-authoring a peer-reviewed research publication**.

AI/ML Engineer — Tech Solutions

Nov 2021 – Jul 2023

- Built **real-time fraud detection systems** using Flask APIs and XGBoost models, enabling low-latency transaction risk scoring and reducing high-risk exposure by **15%**.
- Designed and deployed **scalable backend services and data pipelines** processing **10M+ financial transactions/month**, powering automated risk analysis and decision workflows.
- Developed **production-grade APIs** and asynchronous processing pipelines to support real-time business operations and explainable ML-driven decisions.
- Worked with **AWS infrastructure** (EC2, S3, CloudWatch) and Docker-based deployments to support scalable model serving, monitoring, and reliable backend operations.
- Implemented **CI/CD and MLOps workflows** using Docker, GitHub Actions, and MLflow, improving deployment reliability and reducing iteration time by **30%**.

Education

George Mason University, Fairfax, VA — M.S. Computer Science

2023–2025

GGs Indraprastha University, Delhi, India — B.Tech. Information Technology

2019–2023